

## S6 Supplementary Changelog

*Release history, working-tree additions, and claim-promotion boundaries*

Short title: S6 Changelog - NeuralManifoldDynamics

Robin Langell<sup>1</sup>

<sup>1</sup> *Langell Konsult AB, Vallentuna, Sweden*

Corresponding author: Robin Langell, [hello@noeticdiffusion.com](mailto:hello@noeticdiffusion.com)

Keywords: changelog, measurement contract, MNDM, provenance

### S6.0 Scope and versioning rules

This supplement records changes to the NeuralManifoldDynamics software and data contract, the accompanying manuscript, and the validation evidence used to illustrate that contract. It is not a substitute for a results section and it does not turn a completed pipeline run into an externally replicated scientific finding.

Three evidence boundaries are used throughout:

- **Historical release-bound** means present in commit `ba91e65` (v.2.4, 2026-06-29), documented in the v2.4.0 section of `CHANGELOG.md`.
  - **Current release** means the v3.0.0 package and current contract generation, including the multi-modal, phase-aware, provenance, QC, and gated dynamical-family additions described below.
- The final commit hash must be recorded when the release is frozen.

- **Exploratory, bugfix, or closed** marks work whose scientific interpretation is limited, maintenance-oriented, or explicitly did not survive a stated audit.

The package metadata now reports 3.0.0; the v2.5.0 release history remains named where it is the source of a specific validation or contract claim. The historical v2.4.0 commit remains named where it is the source of a specific validation or contract claim.

## S6.1 Ancestry: MNPS 1.2, 2.1, and 2.3

The older MNPS 1.2-style implementation emphasized a weighted low-dimensional trajectory with limited summary exports. The 2.1 line made the normalization choice explicit by separating subject-anchored and cohort-anchored coordinate layers and recording feature-anchor provenance.

The documented 2.3 line added the Embodied Anchoring Principle as an additive surface (`anchor_state`, `anchor_state_dot`, and `anchor_quality`) aligned to the neural time grid. It also established `event_locked` and `block_native` as derived analysis layers. These layers do not redefine the canonical `mnps_3d = [m, d, e]` chart and are not a fourth canonical axis.

## S6.2 Release-bound v2.4.0: commit **ba91e65**

Relative to HEAD~1 (346b03f, 2026-06-04), the large release commit added the following measurement-contract capabilities:

### S6.2.1 Geometry, support, and provenance

- `standard_invalid_v1` made geometry-validity reporting explicit, including duplicate-sub-coordinate diagnostics and time-grid auditing.
- `anchor_auto_fit` reduced avoidable skips of cohort-anchored coordinates.

- HDF5 outputs gained self-describing anchor, block, row-source, and transform metadata.

The writer supports `anchor_state*`, `anchor_coupling`, `block_windows`, `row_source`, and `features_projection_z`.

### S6.2.2 Embodied and physiological surfaces

- HRV v0.1 added centered time-domain ECG superwindows and the `ecg_hrv_*` quality/coverage columns. Frequency-domain HF and LF/HF metrics are not part of this release surface.
- PPG and pupillometry extraction became available as additive surfaces. Pupillometry remained dataset-dependent and was not validated at scale.
- ECG polarity correction and contamination gating were added to prevent obviously invalid HRV interpretation.

### S6.2.3 Block-native and regional outputs

- `block_native v2` added named window profiles, `block_native_qc.json`, `source_window_index`, CSV/Parquet sidecars, and inter-network `coupl_*` columns.
- Regional fMRI and EEG channel-group trajectories use the same export family, while modality-specific rank and support limits remain explicit.

### S6.2.4 Exploratory MEG path

MEG features are shadow-mapped through the existing 9D contract. Simultaneous MEEG rows carry `row_source` provenance, and `features_projection_z` applies the configured transform before export so physical MEG power does not collapse under raw-space robust-z scaling. The ds003645 pilot comprised five subjects and reported readiness 0.7879; it is an exploratory usable-pilot mapping, not production-scale confirmation.

### S6.2.5 Release-bound validation examples

The release notes report completed exports for ds003838 (130 subjects), ds006036 (88), ds007526 (277 recordings), and dual-anchor reruns for ds003490, ds003506, and ds003509. These are completion or capability counts, not one common inferential sample. For ds003838, the full export contained 27,670 block windows, whereas the corrected stage-statistics run used 26,460 windows and a complete-case analysis sample of  $N = 62$ . The reported listen-mem13 vagal\_index effect ( $d = 1.995$ , Wilcoxon  $p = 1.2 \times 10^{-10}$ ) belongs to the latter statistics package.

## S6.3 Article and supplement changes

The manuscript was expanded from a 2.1-centric contract description into a v2.5.0 methods paper. It now documents:

- the canonical 3D and stratified 9D charts and their fixed projection;
- subject/cohort anchoring and the distinction between feature\_anchors and anchor\_state;
- geometry validity, support-aware Jacobians, regional outputs, and self-describing HDF5 artifacts;
- HRV v0.1, PPG, pupillometry, block-native sidecars, and exploratory MEG;
- the ds003838 internal validation example and the MEG readiness table;
- explicit non-claims about diagnosis, unique biological identifiability, cross-modal homology, and comparative superiority.

S1–S5 are completed as methods, data, QC, clinical-interpretation, and glossary appendices. S2 and S3 preserve explicit unavailable/TBD states where the repository does not contain a locked article-level contrast family. S4 keeps clinical language observational and non-diagnostic.

## S6.4 v2.5.0 release additions

The 2026-07-01 to 2026-08-06 work is promoted into the v2.5.0 release candidate as a cumulative expansion of the v2.4 measurement contract:

- BDF/Figshare infant EEG adapters, privacy-safe metadata, BAD\_ masking, and a repaired connectivity-NaN path;
- Neuropixels Units-to-rate features and DANDI/NWB electrical-series and LFP selection/QC paths, including the one-session DANDI 000458 sensitivity work;
- optional phase-anchor and sleep-EAP extractors, including cardiac/respiratory phase and HEP-related feature surfaces;
- /epoch\_id, stricter row-lineage checks, robust-z degenerate-scale guards, and additional provenance hardening;
- dataset-specific synchronization and pilot configurations for ds004587, ds007216, hypoglycemia/T1D, and related tracks.

The local DANDI and Figshare artifacts support transport, QC, and descriptive claims. They do not establish cross-probe LFP geometry, canonical Neuropixels manifold semantics, or infant behavioral condition effects. The phase-anchor and sleep-EAP paths are optional v2.5.0 capabilities whose interpretation remains dataset- and modality-dependent.

## S6.5 Bug fixes, negative findings, and closed lines

Maintenance work in the post-release tree corrected phase-anchor cardiac sampling, respiratory-index sign handling, degenerate robust-z scaling, sympathetic-index fallback behavior, and BAD\_-masked connectivity. These are correctness fixes, not independent biological findings.

Several lines were intentionally narrowed or closed:

- ds006848 working-memory HRV claims were withheld because 60-second windows overlapped retrieval events; 87.7% of windows carried retrieval contamination.
- The ds003645 pilot showed moderate and bimodal cross-modal convergence; a later holdout rejected matched EEG–MEG residual topographic correspondence.
- The ds003568 regional-MEG source ladder stopped after leakage concerns.
- ds007216 fMRI harmonization and final-test lines remained mixed or closed at the available sample size.
- DANDI 000458 contact/reference comparisons are descriptive one-session sensitivity checks; their independently normalized coordinates are not orientation-identified across runs.

## S6.6 Provenance and release-freeze checklist

Generated HDF5, CSV, Parquet, QC, and manifest files are run products. Diaries and logs document internal reproducibility, but their presence does not make a result peer-reviewed or independently replicated. Diary identifiers are not unique in the current repository; full filenames and run directories should be used.

Before freezing a release commit:

- confirm `__about__`.`__version__` and the matching `CHANGELOG.md` section;
- align the root `README`, manuscript, and supplement version strings;
- create an annotated Git tag or otherwise freeze the cited commit;
- attach every numeric table to a run directory and analysis sample size;
- separate code capability, completed export, smoke test, internal validation, and inferential result;
- retain exploratory and negative findings as such;
- re-run the Typst package with the shared template and bibliography files.

## S6.7 v2.5.0 release notes and Zenodo metadata synchronization

The v2.5.0 release notes now include the detailed description previously kept only as Zenodo metadata. The cumulative release surface is:

### MEG ingest (exploratory/beta)

- `meg_mag_*`, `meg_grad_*`, and `meg_*` feature columns are extracted from Neuromag FIF files through MNE-Python.
- Shadow mapping routes each `meg_*` feature type through the same 9D subcoordinate slot as its `eeg_*` counterpart; the projection machinery is unchanged.
- Simultaneous MEEG recordings carry explicit `row_source/` provenance under `mndm.row_source.v1`, replacing an implicit positional half-split.
- `features_projection_z` applies `log10`, `robust-z`, and clipping before export, which is required for MEG spectral power near  $10^{-2.5}$  W.
- The ds003645 five-subject pilot reached readiness 0.7879 and remains exploratory rather than full-cohort confirmation.

### Embodied anchoring and physiology

- HRV v0.1 exports centered time-domain `ecg_hrv_*` columns, including HR, IBI, SDNN, RMSSD, pNN50, NN count, artifact/coverage fractions, and quality. Optional SampEn and DFA columns are available when configured; HF and LF/HF are not part of the v0.1 surface.
- PPG contributes rate, amplitude, variability, and quality features to `vascular_index`; pupillometry contributes diameter, volatility, blink-rate, and quality features when available.

- ECG polarity correction and HRV contamination flags are exported as measurement-quality controls.
- The optional v2.5 phase-anchor and sleep-EAP surfaces extend this layer with cardiac/respiratory phase, RR/HR, inhale fraction, and HEP-related features.

### **Block-native and geometry contract**

- Block-native v2 exports `block_native_qc.json`, named window profiles, `source_window_index`, Parquet/CSV sidecars, and `coupl_*` inter-network Jacobian columns.
- `anchor_auto_fit` provides one-shot per-run cohort-anchor fitting.
- `standard_invalidty_v1` records duplicate-subcoordinate tolerance, per-subject diagnostics, and time-grid validity.
- Clinical TSV joins and conventional EEG ICU/coma proxy packs are available as additive outputs.
- OpenNeuro download paths use the `uvx openneuro-py@latest` workaround where configured, with the corresponding dependency metadata recorded in the root requirements.

### **Validation and claim ceiling**

The release notes report six additional cohort paths. For ds003838, the full export contains 130 completed subject-level HDF5 files and 27,670 block-native windows; the corrected stage-statistics package uses  $N = 62$  complete cases for the `listen-mem13 vagal_index` result ( $d = 1.995$ , Wilcoxon  $p = 1.2 \times 10^{-10}$ ). The export count and inferential sample must not be conflated.

The ds006848 encoding m/d analyses use  $n = 30$ , while working-memory HRV claims are withheld because 87.7% of 60 s windows overlap retrieval events. DANDI Units/LFP and Figshare

infant EEG outputs remain transport, QC, or descriptive evidence. No v2.5.0 result establishes clinical diagnosis, individual prediction, cross-probe LFP geometry, or independent replication.

The detailed machine-readable release entry is in the root `CHANGELOG.md`.

## **S6.8 v2.6.0 local-dynamics interpretation layer**

Version 2.6.0 adds versioned local-Jacobian metrics and opt-in finite-time response summaries on the unchanged  $\hat{J}$  estimator. The outputs carry coordinate-contract provenance and `validation_level=model_derived`; they are not S3-licensed empirical NDT quantities. The release also defines the boundary for discrete stochastic reachability: opt-in  $W_Q$  may use only the Gate E recording-level transition-residual covariance with declared one-step semantics. It is not  $\hat{a}$ , Itô process noise, or a default family export, and common EEG, fMRI, and ephys profiles leave it disabled. The written `measurement_validity` and `claim_status` fields remain distinct from this model-derived validation label.

## **S6.9 v3.0.0 contract generation and validation boundary**

The v3.0.0 generation keeps the canonical `mnp3d = [m,d,e]`, stratified 9D subcoordinates, and configured projection. It adds the gated family namespace `/dynamical_families/{diffusion,destination,resilience}/v1`; the older `/orthogonal_dynamics/` path is read-only. Common EEG, fMRI, and ephys profiles leave family generation disabled unless a named configuration opts in. Diffusion uses local increment covariance and ingest always exports `drift=None`, so  $A_{bD}$  and  $R_{b\over a}$  remain `not_testable`. Family outputs preserve `computation_status`, `measurement_validity`, `grain`, `provenance`, and `claim_status=no_biological_claim`; windows are not biological replicates. The separate `/support_signature/v1` describes file-level capabilities

and coordinate support. Opt-in discrete  $W_Q$  remains under `/stochastic_reachability/v1` and uses only Gate E recording-level transition-residual covariance, not  $\hat{a}$  or process noise.

The v3 validation evidence has separate scopes. A bounded CHART audit passed the orthogonal and isotropic identities with maximum discrepancy  $2.9 \times 10^{-14}$ ; anisotropic sensitivity and 3D-to-observed-subspace information loss are expected findings. Its scalar diagnostic is unregularized and is not a production C1 gate. The synthetic M3 extension covered six systems with twelve independent training fits and 144 fixed held-out query sources, using shared evaluation-neighbour covariance. The integrated validation suite contained 117 tests, including 18 dedicated extension tests. It reports an estimator bias–variance trade-off rather than a universal winner. The partial-system memory witness was  $6.58 \times 10^{-9}$  and the noisy system witness 0.0441; these are Frobenius norms of model-derived Gaussian conditional-lag diagnostics, not estimator diagnostics, Markov-sufficiency or biological qualifications. The extension uses synthetic 9D systems and does not validate empirical drift, an Itô drift, production ingest C1, or biological interpretation.

The release claim ceiling is therefore: v3.0.0 is a reproducible, gated measurement-contract generation with bounded synthetic coordinate and dynamical-family evidence. It remains separate from empirical biological qualification and from any claim that one local estimator is generally optimal.

## Bibliography

## Competing Interests

The authors have declared that no competing interests exist.